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## ① What is Data Analysis

: Data Analysis is the process of inspecting, cleaning, transforming and modeling data with the goal of discovering useful information, drawing conclusions and supporting decision-making. It is a crucial step before building any machine learning or AI system as it helps us understand the story hidden inside data.

## ② Describe how to plan a Data Science Project, listing all steps

: A Data Science Project Plan is a structured roadmap that defines the sequence of steps to be followed when solving a problem using data. It ensures that efforts are organized, goal-driven and measurable, preventing wasted time on unnecessary tasks.

## All steps for DS project.

### ① Define the problem

! What is the business or research question?

! Example: Predict customer churn for a telecom company

### ② Collect Data

! Identify data sources: CSV files, APIs, databases, sensors, etc.

### ③ Data preprocessing & cleaning

! Handle missing values, duplicates, outliers.

! Transform data into an analyzable format.

### ④ Exploratory Data Analysis (EDA)

! Use descriptive statistics and visualizations to understand data

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⑤ Feature Engineering  
: Create meaningful variables  
features for the model.

⑥ Model Building  
: Apply machine learning algorithms

⑦ Model Evaluation  
: check performance using metrics  
like Accuracy, F1-Score, RMSE

⑧ Deployment  
: Integrate the model into  
real-world system (web app, dashboard)

⑨ Monitoring & Maintenance:  
! Continuously check performance  
and update as data changes.

④ What are Tensor?

: In mathematics, machine learning and computer science, a tensor is a generalization of scalar, vector, and matrices to higher dimensions. It serves as a container for numerical data structured along an arbitrary number of axes.

- Provide an in-depth explanation of tensors with simple example using Numpy

i import numpy as np

Scalar = np.array(5)

Vector = np.array([1, 2, 3])

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```
matrix = np.array([[1,2],[3,4]])
```

```
tensor_3d = np.array([[[  
[1,2],[3,4]],[[5,6],[7,8]  
]])
```

```
print("Scalar:", Scalar)
```

```
print("Vector:", Vector)
```

```
print("Matrix: 'm'", matrix)
```

```
print("3D Tensor: 'm'", tensor_3d)
```